

RAY KWAN

SOFTWARE ENGINEER

Markham, ON ■ (647) 210 3443 ■ lokto.kwan@gmail.com

SUMMARY

Software Engineer with **7+ years of experience** in fast-paced startup environments. Proven ability to drive product success and growth by leading teams that secured contracts with **6+ NHL teams**

WORK EXPERIENCE

SOFTWARE ENGINEER BAUER HOCKEY

Jan 2026 - present

- Evaluated **7 scanning and CNC solutions** and selected one that reduced costs without compromising quality.
- Integrated **Aetrex API** to automate customer data transfer from foot scanning hardware into Skatescribe, delivering recommended skate specs and eliminating manual data entry and operator error.
- Built the foundational cloud infrastructure on **AWS** to scale globally, deploying servers, load balancers, and databases across multiple regions to support a target of **700+ machines** worldwide.

LEAD ENGINEER (promoted from FULL STACK DEV) SKATESCRIBE

May 2021 - present

- Led a cross-functional team of **7 engineers** to drive technical vision for a precision hockey skate tuning system deployed across **6+ NHL teams**, and **3+ AHL affiliates**
- Architected migration from **monolith** to **microservices** by decoupling the order processing system into a dedicated service, increasing operational efficiency by **25%** and improving data accuracy by **70%**.
- Created the **first production-ready prototype of the skate profiling and sharpening system**, integrating a laser profiler with a CNC mill via a Python backend, React UI, and PostgreSQL database.
- Invented proprietary signal processing algorithms for smoothing and joining noisy data, achieving $\pm 1 \mu\text{m}$ precision via least-squares optimization — establishing the technical foundation for all production machines.
- Managed high-availability cloud infrastructure on Linode, deploying containerized environments with **NGINX**, ensuring seamless user experience across **19+ locations**.

FULL STACK DEVELOPER COGNITIVE CENTIVIZER

Sept 2020 - May 2021

- Developed full-stack cognitive assessment games in **React** and **Node.js** for research clients, collecting data across **100+** patients to assess cognitive function

FULL STACK DEVELOPER AUTOCASE

May 2019 - June 2020

- Developed a cost-benefit platform in **AngularJS** and **Python** for government and architecture firms, transforming economic research models into interactive visualizations to inform policy decisions and assess impacts on building occupants and the general population..



SKILLS

Languages & Frameworks

- Python
- Flask
- Javascript
- ReactJS/ NextJS
- AngularJS
- NodeJS
- Java
- PSQL

Libraries

- Numpy/ Scipy
- Pandas
- MUI

DevOps

- Docker
- Linode
- AWS

WEBSITES

- [Website](#)
- [LinkedIn](#)

Projects

- [Music Player](#)
- [Wordle Copycat](#)
- [Graphing Library](#)
- [ATM](#)

EDUCATION

B.Sc. Computer Science

University of Toronto

2016 - 2021

Dean's Honor (**CGPA: 3.72**)

Digital Growth Marketing

Talent Accelerator